

Optimal Smoothing and Monitor Strategies for Dual-Frequency Dual-Constellation (DFDC) LADGNSS under Anomalous Ionospheric Conditions

Gihun Nam, Dongchan Min, Noah Minchan Kim, and Jiyun Lee, *Korea Advanced Institute of Science and Technology (KAIST)*
Sam Pullen, *Stanford University*

ABSTRACT

Dual-Frequency, Dual-Constellation Local-Area DGNSS (DFDC LADGNSS) modernizes and expands upon L1-based LADGPS to support autonomous vehicles with improved system availability and robustness against ionospheric disturbances. Recent research has evaluated several different LADGNSS smoothing strategies using measurements on GPS L1/L5 and Galileo E1/E5a frequencies to determine the optimal approach under nominal conditions. However, under anomalous ionospheric conditions, the optimal smoothing procedure may be different from the one preferred under nominal conditions. In addition, system performance greatly depends on the ionospheric monitoring algorithm when a Single Frequency (SF) or Divergence-Free (DF) smoothing process is used.

To examine this problem, the performance of SF and DF smoothing processes is evaluated by simulation in terms of Maximum Ionosphere-Induced Vertical Errors (MIEVs), which quantify the anomalous ionospheric impact on DFDC LADGNSS. Ionosphere-Free (IF) smoothing, which removes almost all ionospheric delay impacts, is evaluated based on its nominal (H_0) Vertical Protection Level (VPL). A dual frequency ionospheric gradient monitor is also introduced to fully take advantage of multiple-frequency measurements. MIEVs and VPLs are generated for 27-satellite GPS and Galileo constellations to determine which smoothing and monitor strategies give the best performance (lowest MIEVs or VPLs) under anomalous ionospheric conditions.

1.0 INTRODUCTION

Unmanned aerial vehicle (UAV) navigation requires high-performance GNSS positions and velocities as well as outputs from other devices such as inertial and vision sensors. The authors have studied the use of local-area differential GNSS (LADGNSS) to provide the best possible accuracy and integrity to UAVs within 25 - 50 km of a LADGNSS reference station. In [1] and earlier papers, we proposed a LADGNSS reference system based on a simplification of the Ground Based Augmentation System (GBAS) architecture [2] to provide the integrity-monitoring capabilities of GBAS in a smaller, more portable, and less-expensive system that takes advantage of two-way communications between reference and UAVs. A variation of this design has been implemented and tested in the flight tests reported in [1].

Our most recent paper [3] extends this original single-frequency LADGNSS design by making use of measurements on the GPS L5 frequency and the Galileo E1 and E5a frequencies, creating a Dual-Frequency, Dual-Constellation (DFDC) version of LADGNSS. This adds more ranging measurements to GPS L1-only LADGNSS and adds robustness against ionospheric disturbances due to the ability to combine L1/E1 and L5/E5a measurements to remove all or part of the residual (uncorrected) ionospheric errors affecting users. Since several different methods of combining and smoothing these measurements exist, [3] provided an analysis that identifies the best-performing methods as a function of the operational conditions that DFDC LADGNSS will support. However, the analysis in [3] focused on nominal ionospheric conditions where ionospheric gradients do not generate large errors regardless of the processing method.

This paper examines the optimal measurement combination and smoothing approaches that apply to LADGNSS UAV navigation when the possibility of anomalous ionospheric conditions is considered. Single and (future) dual-frequency GBAS has already established means to quantify the threats posed by these conditions (e.g., see [4, 5, 6]). However, GBAS operations are constrained by the precisely defined flight paths of aircraft approaching and landing on runways at large airports, whereas UAVs are likely to fly in all directions around LADGNSS reference stations. Therefore, a new element addressed in this paper is determining the degree

to which actual UAV flight paths can be approximated by the one-dimensional straight-line paths used to examine anomalous ionospheric gradient impacts for GBAS users.

Section 2.0 of this paper reviews the anomalous ionospheric “wave front” threat model geometry that applies to GBAS, expands on it to represent 2-D UAV flight paths, and then demonstrates how one-dimensional analysis can bound errors from 2-D paths based on examining several scenarios. Section 3.0 expands upon existing ionospheric error models to include the impact of multi-frequency smoothing, adds single and dual-frequency monitors, and then identifies a simulation-based procedure to search for the combination of ionospheric front parameters that produces the largest range-domain errors after monitoring. Section 4.0 uses the resulting range-error models and sets of “worst” parameters to evaluate the largest vertical position errors and protection levels, while Section 5.0 summarizes the paper.

2.0 ANOMALOUS IONOSPHERIC IMPACTS ON DFDC LADGNSS

2.1 Moving Ionospheric Front Model

A spatially linear semi-infinite wave front with constant propagation speed was proposed as the ionospheric threat model for GBAS in [4]. Fig 1 illustrates this model, which is parameterized by four threat parameters: the gradient or “slope” of the ramp (g), its width (w), propagation speed of the ionospheric anomaly front relative to the ionospheric pierce point (IPP) of the ground facility (Δv), and a maximum delay difference constraint (D_{max}). The ranges of the threat parameters we used in this study are the same with those of the CONUS threat model [4]. The speed of airborne IPP relative to the ground IPP can be approximated as the speed of airborne, v_{air} , because they are spatially close to each other within the LADGNSS coverage compared to the distance to a satellite. A v_{air} of 15 m/s for UAVs is used in this study while the speed of aircraft for GBAS is set to 70 m/s. x_{air} is the distance of an evaluation point (or target point) to the ground facility.

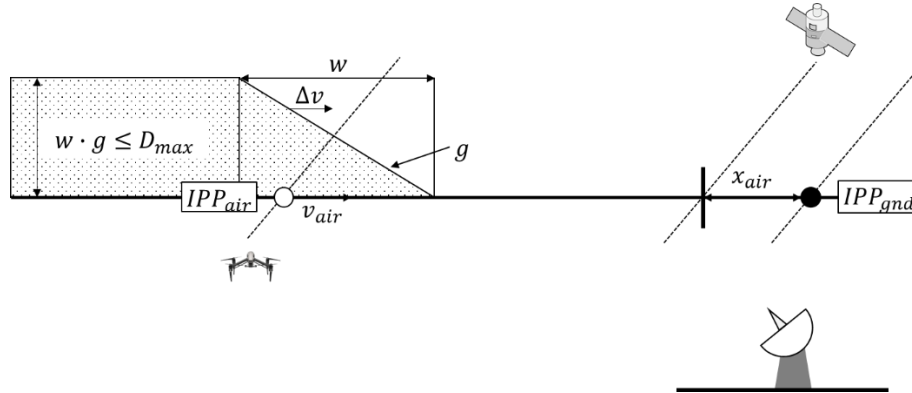


Figure 1: Ionospheric wave-front model (side-view)

2.2 Ionospheric Impact Scenario

Prior work done for GBAS evaluated the effect of ionospheric anomaly front during precision approach operations assuming the aircraft follows a straight path with a constant speed [5, 7, 8]. For LADGNSS, UAVs will move in all directions relative to the ground facility, and thus all flight paths including curved ones should be covered to estimate the largest possible errors at a particular evaluation point. This causes the search procedure of the worst ionosphere-induced differential range error to be complicated. In this study, to address this difficulty, we propose new one-dimensional ionospheric impact scenarios which simplify this search procedure while covering all ‘two-dimensional’ flight paths of UAV with a constant speed. This method classifies flight scenarios in terms of the moving direction of the ionospheric front and UAV net motion relative to the ground facility, assuming a one-dimensional geometry and a constant UAV speed. LADGNSS utilizes Code-Carrier Divergence (CCD) monitors which can detect ionospheric anomalies by monitoring code-carrier divergence rates at the ground facility and airborne. The detection capability of the CCD monitor depends on the relative speed of ionospheric front with respect to each IPP. While the effect of adding the airborne CCD monitor is small compared to the ground CCD monitor, it is difficult to address the airborne CCD monitor whose divergence rate

varies with the airborne speed. Thus, this study chooses a conservative approach by assuming that the airborne CCD monitor is not used.

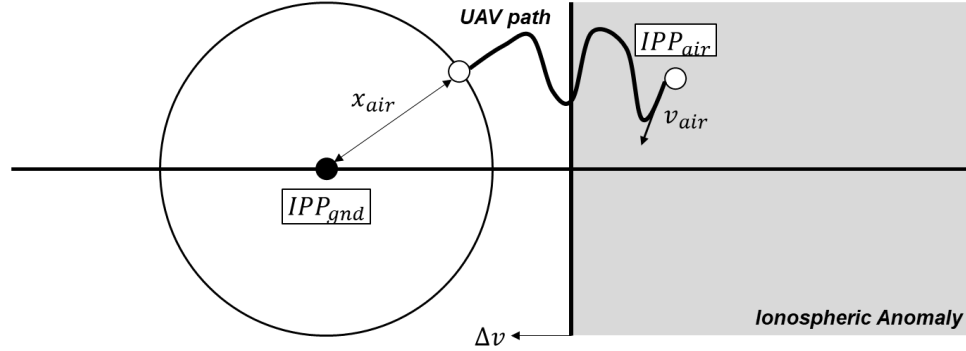


Figure 2: An example of arbitrary flight path operations with a constant UAV speed under anomalous ionospheric conditions (top-view). x_{air} denotes the distance between the ground facility and target point, v_{air} is a constant UAV speed, and Δv is a speed of the ionospheric anomaly front relative to the IPP of the ground facility.

Below, we show that the worst ionosphere-induced differential range error of the proposed scenarios modeled with a one-dimensional geometry and a constant UAV speed can overbound that of all two-dimensional scenarios with an arbitrary UAV flight path flying at constant speed. Fig 2 shows an example of arbitrary flight directions with a constant speed under anomalous ionospheric conditions. UAV motion under the ionospheric anomaly front, can be decomposed into the two components; perpendicular or parallel to the ionospheric front. The motion parallel to the ionospheric front would not experience the change of ionospheric delays, and Code-Carrier-Divergence (CCD) is induced only by the motion orthogonally projected onto the normal vector of the ionospheric front. Thus, we simplify the two-dimensional scenario by expressing it as one-dimension impact scenario with a varying speed.

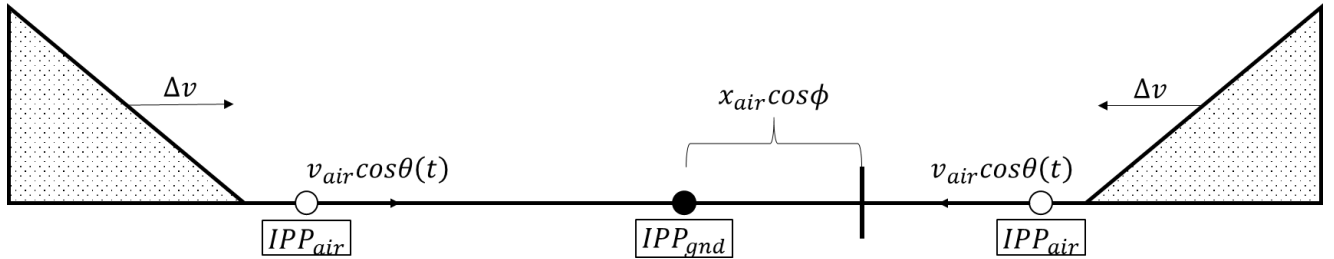


Figure 3: Four different ‘one-dimensional’ scenarios modeled by projecting curved flight path operations with a constant UAV speed (v_{air}) to a line perpendicular to the ionospheric front (side-view). They are classified in terms of the moving direction of ionospheric anomaly front and direction of UAV net motion. $\theta(t)$ denotes the angle between the velocity vector of ionospheric front and velocity vector of UAV at time t , and ϕ is the angle between the normal vector of ionospheric anomaly front and vector passing through the ground IPP and airborne IPP.

Table 1: Four scenarios whose target point is located to the right of the ground facility

	Moving Direction of Ionospheric Anomaly	Direction of UAV Net Motion
First Scenario	Right to Left	Right to Left
Second Scenario	Left to Right	Right to Left
Third Scenario	Right to Left	Left to Right
Fourth Scenario	Left to Right	Left to Right

Fig 3 and Table 1 describe four different ‘one-dimensional’ scenarios classified in terms of 1) the moving direction of ionospheric anomaly front, 2) direction of UAV net motion from a starting point to a target point. We consider the scenarios whose target points are located at the right side of the ground facility only, since those with target points at the left side are symmetric to the original four scenarios. Thus, only four scenarios will be simulated in Section 3 to determine the worst ionosphere-induced differential range error. The rest of this subsection prove that the first and third scenarios with a constant UAV speed always bound that of a varying UAV

speed for the first scenario, and thus the worst range-domain error simulations can be further simplified with an assumption of a constant UAV speed. The same assumption can be applied to the other three scenarios.

We compare the first scenario with a varying speed (denoted as ‘Curved Case’), which was originally a curved path operation with a constant UAV speed, to the first and third scenarios with the same UAV speed (denoted as ‘First Case’ and ‘Second Case’ in Fig 4., respectively). The worst initial geometry of the ionospheric anomaly front that makes the ‘Curved Case’ to have the worst differential range error at the target point can be found by changing the starting point of the ionospheric front for a given UAV path. Other threat model parameters (g , w and Δv) are assumed to be fixed. Let us consider First and Second Cases which have the same worst initial geometry with Curved Case except the starting point of the UAV. The UAVs of First and Second Cases start farther away from the target point than the UAV of Curved Case, thus the airborne IPPs of three operations reach the target point at the same time. The relative motion of the ionospheric front with respect to the ground facility are identical for three cases.

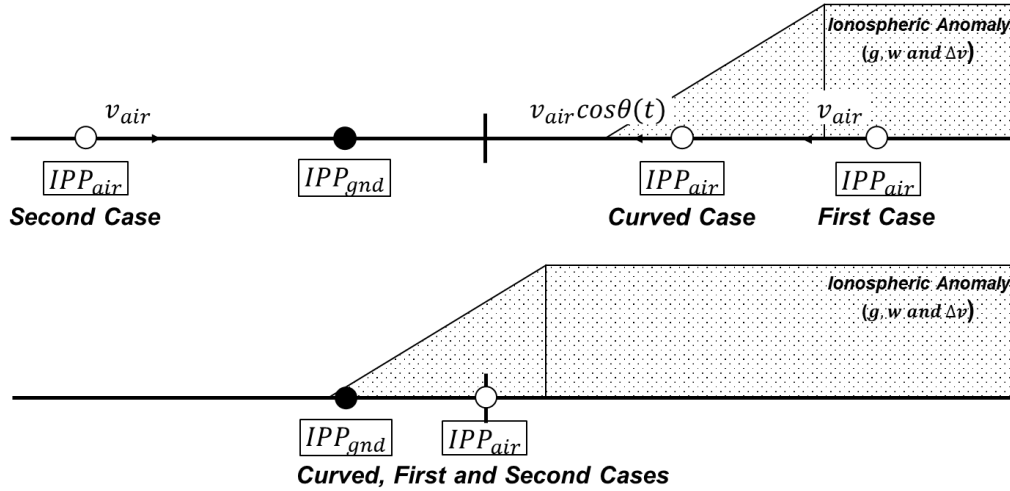


Figure 4: First scenario with a varying speed (Curved Case) and the first and third scenarios with a constant speed (First and Second Cases) in a side-view. The initial and final geometry of the operations are shown in the top and bottom plots, respectively.

With the conditions of the initial (top plot in Fig 4) and final (bottom plot in Fig 4) geometries of Curved, First and Second Cases, the followings are always true: 1) the change of ionospheric delays experienced from the starting point to the target point by the UAV of First Case is larger than or equal to that of Curved Case, 2) the change of ionospheric delays experienced from the starting point to the target point by UAV of Second Case is less than or equal to that of Curved Case, and 3) the ionospheric spatial gradients when UAV reaches at the target point are the same for Curved, First and Second Cases. This thus tells us the smoothed ionospheric delays of airborne (C_{air}) for three operations satisfy the following inequality at the target point.

$$C_{air,first} \geq C_{air,curved} \geq C_{air,second} \quad (1)$$

Three operations have the same smoothed ionospheric delays of ground facility (C_{gnd}) due to identical relative motion of ionospheric front with respect to the ground facility. Thus, the following inequality is also true.

$$C_{air,first} - C_{gnd} \geq C_{air,curved} - C_{gnd} \geq C_{air,second} - C_{gnd} \quad (2)$$

The absolute value of second term in Eq. (2) represents the worst ionosphere-induced differential range error of Curved Case ($\epsilon_{worst,curved}$), and it is bounded by the maximum value of two differential range errors for the First and Second Cases.

$$\epsilon_{worst,curved} = |C_{air,curved} - C_{gnd}| \leq \max\{|C_{air,first} - C_{gnd}|, |C_{air,second} - C_{gnd}|\} \quad (3)$$

Therefore, there are always differential range errors of the first and third scenarios with a constant speed, which can bound the worst ionosphere-induced differential range error of the first scenario with a varying speed. Thus, the worst ionosphere-induced differential range error for any first scenario with a varying speed can be covered by considering those of the first and third scenarios with a constant speed.

3.0 IONOSPHERE-INDUCED ERRORS IN THE RANGE DOMAIN

This section describes how ionosphere-induced errors in the range domain are calculated using the ionospheric moving front model and impact scenarios in Section 2. The code-carrier smoothing methods and CCD monitor have to be addressed to determine the worst ionosphere-induced differential range error. LADGNSS applies a code-carrier smoothing technique to reduce the multipath and thermal noise errors in raw code measurements at both the ground facility and airborne. Several different smoothing methods are available using dual frequency measurements. For the DFDC LADGNSS system, we can consider three different smoothing methods; Single Frequency (SF), Divergence-Free (DF), and Ionosphere-Free (IF). The worst-case errors are computed for SF and DF only, because IF can remove the ionosphere-induced range errors even under anomalous ionospheric conditions. The CCD monitor detects the satellite failure or ionospheric anomaly by observing the code-carrier divergence rate. LADGNSS utilizes the CCD monitors at both the ground facility and airborne, but the ground CCD monitor only is considered in this study as addressed in Section 2.

This study conducts the exhaustive search algorithm described in [7] to compute the worst ionosphere-induced differential range error. The algorithm examines all geometries and permutations of threat parameters for a given ionospheric impact scenario, and stores the smoothed ionospheric delay and the test statistic of the CCD monitor for each case. It then searches the worst differential range error at a given target point among the undetected cases whose test statistic of the CCD monitor do not exceed its minimum detectable divergence rate (MDDR) anytime during the UAV operation.

The exhaustive search based on the recursive approach which updates the smoothed ionospheric delay and test statistic of the CCD monitor at each epoch is computationally inefficient. Thus, we introduce explicit expressions of smoothed ionospheric delays and the test statistic of the CCD monitor to compute undetected differential range errors. The expression for SF [7] is summarized and that for DF is newly derived in the following subsections.

3.1 Range Error Expression with Carrier-Smoothing Code

1. Single Frequency Smoothing Method

The x -frequency-based single frequency (SF) smoothed code measurement, $\widehat{\rho_{x,SF}}(t)$, at the current epoch, t , is expressed as follows:

$$\widehat{\rho_{x,SF}}(t) = \frac{T}{\tau} \rho_x(t) + \frac{\tau - T}{\tau} \{ \widehat{\rho_{x,SF}}(t - T) + \varphi_x(t) - \varphi_x(t - T) \} \quad (4)$$

where $\rho_x(t)$ is the frequency x raw code measurement, $\varphi_x(t)$ is the frequency x carrier phase measurement, τ is the smoothing time constant, and T is the GNSS measurement sampling time.

The explicit expression of the SF smoothed ionospheric delay is derived in [7] using the ionospheric threat model explained in Section 2. The SF smoothed ionospheric delays for the ramp inputs of increasing and decreasing delays denoted as $C_{SF,inc}(t, g_t, T_w)$ and $C_{SF,dec}(t, g_t, T_w)$, respectively, are as follows [7]:

$$C_{SF,inc}(t, g_t, T_w) = \begin{cases} 0, & t < 0 \\ g_t \left\{ t + 2\tau \left(e^{-\frac{t}{\tau}} - 1 \right) \right\}, & 0 \leq t < T_w \\ g_t \left\{ T_w - 2\tau e^{-\frac{t}{\tau}} \left(e^{-\frac{T_w}{\tau}} - 1 \right) \right\}, & T_w \leq t \end{cases} \quad (5)$$

$$C_{SF,dec}(t, g_t, T_w) = \begin{cases} g_t T_w, & t \leq 0 \\ g_t \left\{ T_w - t - 2\tau \left(e^{-\frac{t}{\tau}} - 1 \right) \right\}, & 0 \leq t < T_w \\ g_t \left\{ 2\tau e^{-\frac{t}{\tau}} \left(e^{-\frac{T_w}{\tau}} - 1 \right) \right\}, & T_w \leq t \end{cases}$$

where g_t is the temporal gradient, and T_w is the time required for the ionospheric front to completely pass through the relevant ground or airborne IPP. Thus, g_t and T_w depend on the ionospheric impact scenario.

2. Divergence-Free Smoothing Method

The x -frequency-based divergence-free (DF) smoothed code measurement, $\widehat{\rho_{x,DF}}(t)$, at the current epoch, t , is expressed as follows:

$$\widehat{\rho_{x,DF}}(t) = \frac{T}{\tau} \rho_x(t) + \frac{\tau - T}{\tau} \left\{ \widehat{\rho_{x,DF}}(t - T) - \frac{f_y^2 + f_x^2}{f_y^2 - f_x^2} \cdot (\varphi_x(t) - \varphi_x(t - T)) + \frac{2f_y^2}{f_y^2 - f_x^2} \cdot (\varphi_y(t) - \varphi_y(t - T)) \right\} \quad (6)$$

where f_x is the frequency x .

The DF smoothed ionospheric delays for the ramp inputs of increasing and decreasing delays denoted as $C_{DF,inc}(t, g_t, T_w)$ and $C_{DF,dec}(t, g_t, T_w)$, respectively, are as follows:

$$C_{DF,inc}(t, g_t, T_w) = \begin{cases} 0, & t \leq 0 \\ g_t t, & 0 \leq t < T_w \\ g_t T_w, & T_w \leq t \end{cases} \quad (7)$$

$$C_{DF,dec}(t, g_t, T_w) = \begin{cases} g_t T_w, & t \leq 0 \\ g_t (T_w - t), & 0 \leq t < T_w \\ g_t T_w, & T_w \leq t \end{cases}$$

Eq. (7) is derived from Eq. (6), and its derivation is given in Appendix A.

Based on Eq. (5) and Eq. (7), the differential range error can be computed for each scenario. The first scenario is classified into two categories in terms of the speed of an ionospheric front relative to an airborne IPP: *slow-moving* and *fast-moving*. The propagation speed of a *slow-moving* front is slower than that of the airborne IPP with respect to the ground IPP, and a *fast-moving* front is the opposite. In the case of the *slow-moving* scenario, the starting point of the airborne IPP is located at the trailing edge of the wave in order to ensure that the airborne IPP experiences the entire ionospheric delay change in the simulations. For the same reason, the starting point of the airborne IPP is located at the leading edge of the wave in the case of the *fast-moving* scenario. Fig 5 illustrates these initial geometries of the *slow-* and *fast-moving* scenarios.

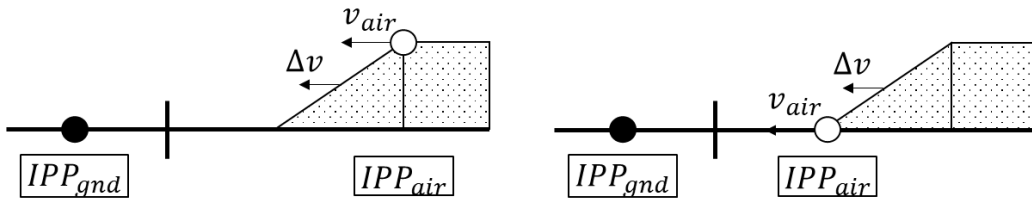


Figure 5: The initial geometry of the *slow-moving* (left) and *fast-moving* (right) scenario.

The differential range errors for the *slow-* and *fast-moving* fronts of the first scenario denoted as $\varepsilon_{slow}(t)$ and $\varepsilon_{fast}(t)$, respectively, are then calculated as follows:

$$\varepsilon_{slow}(t) = C_{dec}(t, g_{air}, T_{air}) - C_{inc}(t - t_{gnd}, g_{gnd}, T_{gnd}) \quad (8)$$

$$\varepsilon_{fast}(t) = C_{inc}(t, g_{air}, T_{air}) - C_{inc}(t - t_{gnd}, g_{gnd}, T_{gnd})$$

where the parameters with an initial separation of D between the ionospheric front and the ground IPP are:

$$g_{air} = g|\Delta v - v_{air}|, \quad g_{gnd} = g\Delta v \quad (9)$$

$$T_{air} = \frac{w}{|\Delta v - v_{air}|}, \quad T_{gnd} = \frac{w}{\Delta v}, \quad t_{gnd} = \frac{D}{\Delta v}$$

Finally, the differential range error when the airborne arrives at the target point can be computed by substituting t in Eq. (8) with the following times.

$$\begin{aligned} t_{TP}^{slow} &= \frac{D + w - x_{air}}{v_{air}} \\ t_{TP}^{fast} &= \frac{D - x_{air}}{v_{air}} \end{aligned} \quad (10)$$

3.2 Code-Carrier Divergence Monitor

The expression for the test statistic of the CCD monitor for increasing and decreasing ionospheric delays, $Z_{inc}(t, g_t, T_w)$ and $Z_{dec}(t, g_t, T_w)$, were presented in [7], and the corresponding equations are as follows:

$$Z_{inc}(t) = 2g_t \left\{ \left(1 - \frac{t + \tau_{ccd}}{\tau_{ccd}} e^{-\frac{t}{\tau_{ccd}}} \right) u(t) - \left(1 - \frac{t - T_w + \tau_{ccd}}{\tau_{ccd}} e^{-\frac{-t+T_w}{\tau_{ccd}}} \right) u(t - T_w) \right\} \quad (11)$$

$$Z_{dec}(t) = -Z_{inc}(t)$$

where τ_{ccd} is the smoothing time constant for a series of two cascaded first-order low-pass filters, and $u(\cdot)$ represents a step function.

The test statistic in Eq. (11) is used to determine the undetected ionospheric anomaly conditions by comparing it with the minimum detectable divergence rate (MDDR). The nominal CCD test statistic is given as normally distributed with a zero mean. Thus, MDDR which is used to ensure a sufficiently low probability of missed detection can be defined as follows [7]:

$$MDDR_{ccd} = (k_{ffa} + k_{md})\sigma_d \quad (12)$$

where k_{ffa} is the multiplier satisfying the allocated fault-free detection requirement, k_{md} is the multiplier satisfying the allocated missed detection requirement, and σ_d is the standard deviation of CCD monitor test statistic under the nominal condition.

In this study, we use the value of σ_d as $0.00399m/s$, which is dominated by the contribution of nominal code-carrier divergence [9]. The requirements of fault-free detection and missed detection are 1×10^{-7} per 15 seconds and 1×10^{-4} per approach, respectively.

3.3 Exhaustive Search for the Largest Undetected Ionosphere-Induced Differential Range Error

The exhaustive search is conducted to compute the worst ionosphere-induced differential range error of operation mode candidates. For the DFDC LADGNSS system, the operation mode candidates listed in Table 2 are considered. The ionosphere-induced differential range error can occur as a combination of the physical separation, x_{air} , and the synthetic separation due to the memory of the gradient remaining in the code-carrier smoothing filter [7]. Among these, the contribution of synthetic separation increases as aircraft speed increases when the ground and airborne use the same smoothing time constant. Because of the low UAV airborne speed of $15m/s$ used in this study, the differential range error for LADGNSS is insensitive to the smoothing time constants. Thus, we consider the candidate with 30-second ground and 30-second airborne smoothing time constants only, as this was selected as the optimal smoothing time constants in [3]. The physical separations (x_{air}) between $1km$ and $25km$ with a step size of $1km$ are considered to cover all operations within primary LADGNSS coverage. The ranges of the parameters and step sizes used in the exhaustive search are listed in Table 3.

Table 2: DFDC LADGNSS system operation candidates

Smoothing Method	Smoothing Time Constants
L1/E1 based Single Frequency Smoothing	$(\tau_{gnd}, \tau_{air}) = (30s, 30s)$
L1/E1 based Divergence-Free Smoothing	
L5/E5a based Single Frequency Smoothing	
L5/E5a based Divergence-Free Smoothing	

Table 3: Threat parameter ranges and step size for simulation

Threat Parameter	Range [Unit]	Step Size [Unit]
g	0 – 500 [mm/km]	5 [mm/km]
Δv	0 – 750 [m/s]	1 [m/s]
w	25 – 200 [km]	25 [km]
D	0 – 1000 [km]	1 [m]

Based on the constructed 2-D table of the worst ionosphere-induced differential range errors for different slopes (g) and relative speeds of the ionospheric front (Δv), we compute the worst vertical position errors in the next following section to evaluate the performance of smoothing strategies under the anomalous ionospheric conditions.

3.4 Dual Frequency Ionospheric Gradient Monitor

The performance of LADGNSS under anomalous ionospheric conditions depends greatly on the monitoring algorithms when SF-based or DF-based smoothing is used, as a significant ionosphere-induced differential range error remains. In addition to the ground and airborne CCD monitors, dual frequency-based monitors are being developed and evaluated to fully take advantage of multiple-frequency measurements ([6], [10]). In this study, we use the “dual-frequency ionospheric gradient monitor” originally proposed in [6] and recently updated in [11]. The basic concept of this monitor is computing separate ionospheric delay estimates for ground and airborne receivers using dual-frequency measurements and then comparing them. The method is summarized in Eqs. 13 – 16 below.

The ionospheric delay estimates of the n^{th} satellite for the ground and airborne are denoted as $\tilde{I}_{gnd,n}$ and $\tilde{I}_{air,n}$, respectively. They are calculated as follows based on the first-order refraction model:

$$\begin{aligned}\tilde{I}_{gnd,n} &= \frac{f_y^2}{f_x^2 - f_y^2} [(PRC_{y,n} + \Delta t \cdot RRC_{y,n}) - (PRC_{x,n} + \Delta t \cdot RRC_{x,n})] \\ \tilde{I}_{air,n} &= \frac{f_y^2}{f_x^2 - f_y^2} (\widehat{\rho_{air,y}^n} - \widehat{\rho_{air,x}^n})\end{aligned}\tag{13}$$

where $PRC_{x,n}$ is the ground-computed differential correction for the frequency x code measurement of n^{th} satellite, $RRC_{x,n}$ is the rate of the frequency x code measurement correction, Δt is the time difference between the current time of measurement and the time when corrections are generated, and $\widehat{\rho_{air,x}^n}$ is the airborne smoothed frequency x code measurement of the n^{th} satellite.

Using the results of Eq. (13), the differential ionospheric delay estimate, $\tilde{I}_{test,n}$, is defined as follows:

$$\tilde{I}_{test,n} = \tilde{I}_{gnd,n} + \tilde{I}_{air,n}\tag{14}$$

Note that the two ionospheric delay estimates are added instead of subtracted because the pseudorange corrections used to generate $\tilde{I}_{gnd,n}$ have the opposite sign from those of the code measurements used to generate $\tilde{I}_{air,n}$. However, the result of Eq. (14) cannot be used as the test statistic because there is a bias introduced by satellite and receiver instrumental errors which are different for each frequency, constellation, satellite and receiver under nominal conditions. To remove the majority of this bias, the test statistic, $I_{test,n}$, is defined as follows by removing the median of differential ionospheric delay estimates for each constellation [11].

$$I_{test,n} = \tilde{I}_{test,n} - \text{median}_{i \in S_c}(\tilde{I}_{test,i})\tag{15}$$

where $\text{median}_{i \in S_c}(\tilde{I}_{test,i})$ represents the median of $\tilde{I}_{test,i}$, which is defined for each set of airborne satellites per constellation.

When rewriting Eq. (15) using the simplified signal error models, the test statistics with odd and even numbers of satellites are denoted as $I_{test,n,odd}$ and $I_{test,n,even}$, respectively, and can be written as:

$$\begin{aligned}
I_{test,n,odd} &= \Delta I_x^n - \Delta I_x^{\frac{N_{air}-1}{2}} \\
&\quad + \frac{f_y^2}{f_x^2 - f_y^2} \left(E_{gnd}^n + E_{air}^n - E_{gnd}^{\frac{N_{air}-1}{2}} - E_{air}^{\frac{N_{air}-1}{2}} + DGD_{gnd}^n + DGD_{air}^n - DGD_{gnd}^{\frac{N_{air}-1}{2}} - DGD_{air}^{\frac{N_{air}-1}{2}} \right) \\
I_{test,n,even} &= \Delta I_x^n - \left(\frac{\Delta I_x^{\frac{N_{air}}{2}} + \Delta I_x^{\frac{N_{air}+1}{2}}}{2} \right) \\
&\quad + \frac{f_2^2}{f_1^2 - f_2^2} \left(E_{gnd}^n + E_{air}^n - \left(\frac{E_{gnd}^{\frac{N_{air}}{2}} + E_{air}^{\frac{N_{air}}{2}} + E_{gnd}^{\frac{N_{air}+1}{2}} + E_{air}^{\frac{N_{air}+1}{2}}}{2} \right) + DGD_{gnd}^n + DGD_{air}^n \right. \\
&\quad \left. - \left(\frac{DGD_{gnd}^{\frac{N_{air}}{2}} + DGD_{air}^{\frac{N_{air}}{2}} + DGD_{gnd}^{\frac{N_{air}+1}{2}} + DGD_{air}^{\frac{N_{air}+1}{2}}}{2} \right) \right)
\end{aligned} \tag{16}$$

where ΔI_x^i is the differential ionospheric delay between ground and airborne for frequency x on the i^{th} satellite, E_{gnd}^i is the contribution of ground multipath and thermal noise on the i^{th} satellite, E_{air}^i is the contribution of airborne multipath and thermal noise, DGD_{gnd}^i is the differential ground receiver antenna group delay on the i^{th} satellite, DGD_{air}^i is the differential airborne receiver antenna group delay, and N_{air} is the number of all-in-view satellites in the airborne receiver. The derivation of Eq. (16) is presented in Appendix B.

Although the median of the differential ionospheric delay estimations is removed, the mean of the test statistic is not zero yet due to the receiver antenna group delay (AGD) under nominal conditions. The AGD is directly affected by the pattern uniformity of the antenna, and thus it shows variation with the azimuth and elevation of the satellite. The absolute AGD can be deterministically estimated by measuring the pattern uniformity for each frequency in an anechoic chamber for all azimuth and elevation angles [12]. In this study, it is assumed that the test statistic can guarantee a zero mean by correcting the raw code measurement based on the corresponding AGD estimation, and the residual error of correction is considered to be included in the multipath error model. Then the minimum detectable differential ionospheric delay (MDID) of n^{th} satellite, $MDID_n$, which is used to ensure a sufficiently low probability of missed detection, can be expressed as follows:

$$MDID_n = (k_{ffa} + k_{md})\sigma_{test,n} \tag{17}$$

where k_{ffa} is the multiplier satisfying the sub-allocated fault-free detection requirement, k_{md} is the multiplier satisfying the sub-allocated missed detection requirement, and the standard deviations of test statistics with odd and even numbers of satellites are denoted as $\sigma_{test,n,odd}$ and $\sigma_{test,n,even}$, respectively, which can be calculated as follows based on Eq. (16):

$$\begin{aligned}
\sigma_{test,n,odd}^2 &= \sigma_{iono,n}^2 + \sigma_{iono,\frac{N_{air}-1}{2}}^2 + \left(\frac{f_y^2}{f_x^2 - f_y^2} \right)^2 \left(\sigma_{gnd,n}^2 + \sigma_{air,n}^2 + \sigma_{gnd,\frac{N_{air}-1}{2}}^2 + \sigma_{air,\frac{N_{air}-1}{2}}^2 \right) \\
\sigma_{test,n,even}^2 &= \sigma_{iono,n}^2 + \frac{\sigma_{iono,\frac{N_{air}}{2}}^2}{4} + \frac{\sigma_{iono,\frac{N_{air}+1}{2}}^2}{4} \\
&\quad + \left(\frac{f_y^2}{f_x^2 - f_y^2} \right)^2 \left(\sigma_{gnd,n}^2 + \sigma_{air,n}^2 + \frac{\sigma_{gnd,\frac{N_{air}}{2}}^2 + \sigma_{air,\frac{N_{air}}{2}}^2 + \sigma_{gnd,\frac{N_{air}+1}{2}}^2 + \sigma_{air,\frac{N_{air}+1}{2}}^2}{4} \right)
\end{aligned} \tag{18}$$

where $\sigma_{iono,i}$ is the standard deviation of the ionospheric decorrelation for i^{th} satellite, and $\sigma_{gnd,i}$ and $\sigma_{air,i}$ are the standard deviations of E_{gnd}^i and E_{air}^i , respectively, in Eq. (16).

The characteristics of ground and airborne multipath and thermal noise greatly affect the performance of this monitor. In order to avoid unnecessary conservatism in evaluating its performance which comes from considering dual frequency multipath errors to be independent, inter-frequency correlation should be addressed. The standard deviation of the n^{th} satellite ground or airborne multipath and thermal noise contribution with a constant inter-frequency multipath correlation coefficient is decomposed as shown in Eq. (19):

$$\sigma_{gnd\ or\ air,n}^2 = \sigma_{gnd\ or\ air,MP,x}^2(\theta_n) + \sigma_{gnd\ or\ air,n,x}^2(\theta_n) + \sigma_{gnd\ or\ air,MP,y}^2(\theta_n) + \sigma_{gnd\ or\ air,n,y}^2(\theta_n) - 2\rho_{IFC}\sigma_{gnd\ or\ air,MP,x}\sigma_{gnd\ or\ air,MP,y} \quad (19)$$

where $\sigma_{gnd\ or\ air,MP,x}$ is the standard deviation of ground or airborne multipath error on frequency x , $\sigma_{gnd\ or\ air,n,fi}$ is the standard deviation of the ground or airborne thermal noise on frequency x , ρ_{IFC} is the correlation coefficient between multipath errors on frequencies x and y , and θ_n is the elevation angle of the n^{th} satellite.

The detection capability of the ionospheric gradient monitor also depends on satellite geometry. Thus, it is addressed in the simulation for computing the maximum ionosphere-induced vertical error (MIEV) described in the following section.

4.0 IONOSPHERE-INDUCED ERRORS IN THE VERTICAL ERROR DOMAIN

4.1 Satellite Impact Scenario

The CCD monitor within GBAS or LADGNSS continuously monitors the code-carrier divergence rate to detect satellite-generated divergence or ionospheric anomalies. The CCD monitor's ionospheric detection capability depends on the relative motion between an observing IPP and the ionospheric anomaly location. If the projected speed of one satellite's IPP (the i^{th} satellite in Fig 6) onto the vector normal to the ionospheric front ($v_{i,proj}$) is close to that of the ionospheric front (v_{front}), the detection capability of CCD monitor against that satellite is greatly degraded, as shown in Fig 6. If the speed match is close enough to make the condition undetectable, the system would suffer from the ionosphere-induced vertical error (IEV) due to the undetected range error on the i^{th} satellite.

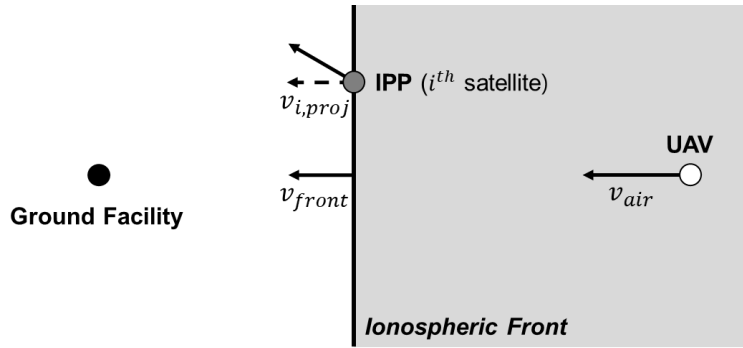


Figure 6: An example satellite geometry causing undetectable range errors.

An undetectable condition like the one above can occur for multiple satellites simultaneously. However, the prior probability of such an event is dramatically reduced as the number of missed detected satellites increases. Previous work [13] investigated actual geometries for three airports (Memphis, Los Angeles/LAX, and New York/JFK) and showed that missed detection of fronts simultaneously affecting more than two satellites in a near-worst-case manner is extremely improbable. Therefore, in this study, the worst-case scenario in which up to two satellites are not detected is considered.

The IEV can be calculated for any satellite or any satellite pair at each epoch. Among these IEVs for each scenario, the maximum IEV (MIEV) over all possible impacted satellite combinations is determined. MIEVs are calculated for one satellite and two satellites impacted scenarios using the following equations [5].

$$MIEV_{1SV} = \max_i \{IEV_i\} = \max_i \{|s_{vert,i}\epsilon_i|\} \quad (20)$$

$$MIEV_{2SV} = \max_{i,j} \{IEV_{i,j}\} = \max_{i,j} \{|s_{vert,i}\varepsilon_i| + |s_{vert,j}\varepsilon_j|\}$$

where $MIEV_{1SV}$ is the MIEV for the one-satellite-impacted scenario, $MIEV_{2SV}$ is the MIEV for the two-satellites-impacted scenario, $s_{vert,i}$ is the element of the weighted least square estimator projecting the i^{th} measurement into vertical position, and ε_i is the undetected range error on the i^{th} satellite.

The undetected range error in Eq. (20) can be determined based on an GNSS satellite almanac and exhaustive search result based on the ionospheric anomaly models in Section 3. At each simulated epoch or point in time, GNSS geometries at one or more ground locations are determined, from which IPP locations and velocities for ground and airborne receivers near that location are calculated (given an assumed airborne flight path relative to the ground receiver location). Maximum undetected range errors given worst-case (hypothetical) ionospheric anomalies are evaluated by looping through all possible ionospheric impact scenarios.

Undetected range errors for each impact scenario are determined as shown in [5]. The one-satellite-impacted scenario assumes the speed of ionospheric front is the same as that of the impacted satellite's IPP. Thus, the undetected range error is determined from an exhaustive search result with zero relative speed of the ionospheric front. For the two-satellites-impacted scenario, it is hypothesized that the ionospheric front is parallel to the line passing through the two impacted satellites' IPPs, and the speed of the ionospheric front is the same as that of projected speed of one IPP to the normal vector of ionospheric front (this is repeated for the 2nd IPP so that all impact cases are considered).

4.2 MIEV Simulation

1. Simulation Method and Conditions

Simulation is also conducted to compute MIEVs (i.e., maximum undetected vertical position errors) and evaluate the performance of the dual frequency ionospheric gradient monitor and each operational mode. In this simulation, IF smoothing is also considered along with the operational mode candidates in Table 2. We compare the simulated Vertical Protection Levels (VPLs) of IF smoothing with MIEVs of the DF and SF smoothing processes because IF removes ionosphere-induced error and is thus unaffected by ionospheric anomalies (to first order). VPLs are calculated using Eq. (21) and Eq. (22) [2]:

$$VPL_{H0} = K_{ffmd} \sqrt{\sum_{i=1}^N s_{vert,i}^2 \cdot \sigma_i^2} \quad (21)$$

$$VPL_{eph} = \max_k (VPL_{eph,k}) \quad (22)$$

$$VPL_{eph,k} = |s_{vert,k}| x_{air} P_k + K_{md} \sqrt{\sum_{i=1}^N s_{vert,i}^2 \cdot \sigma_i^2}$$

where VPL_{H0} is the VPL under the fault-free hypothesis, K_{ffmd} is the multiplier to meet the allocated integrity risk requirement, $s_{vert,i}$ is the element of the weighted least square estimator that projects the i^{th} range measurement into vertical position, σ_i is the standard deviation of a zero-mean Gaussian distribution bounding all range errors of the i^{th} corrected code measurement in the tails, $VPL_{eph,k}$ is the VPL under an assumed ephemeris fault on the k^{th} satellite, x_{air} is the distance between (ground) reference receiver centroid and user (airborne) receiver, P_k is the ephemeris error decorrelation parameter for the k^{th} satellite ($P_k = 0.00018$ is used in this study based on [14]), and K_{md} is the multiplier to satisfy the allocated missed detection probability of an ephemeris fault. The final VPL applied by the user is the maximum between VPL_{H0} and VPL_{eph} . Note that VPL_{H1} , or the VPL under the hypothesis of a single reference receiver failure (out of 2 or more) is not included in the simulation because it depends upon real-time differences between reference receiver measurements which are not simulated because they generally do not affect the final VPL unless ground receiver failures are present.

The dual frequency ionospheric gradient monitor cannot be addressed in the exhaustive search for the worst ionosphere-induced differential range error because its performance depends upon satellite geometry. Therefore, the performance of ionospheric gradient monitor is evaluated using the MIEV calculation process shown in Fig 7.

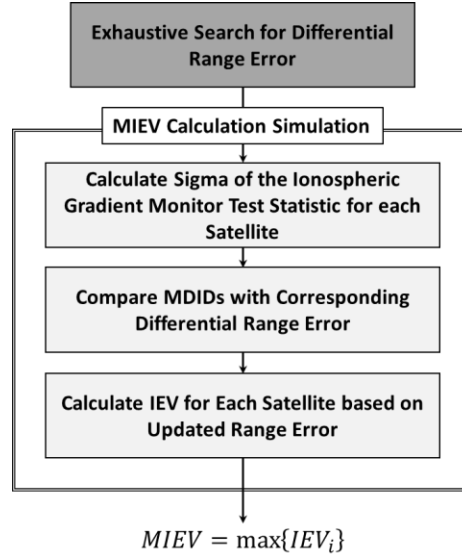


Figure 7: MIEV calculation procedure for the dual frequency ionospheric gradient monitor

The starting point of this procedure is the worst ionosphere-induced differential range error from the previous exhaustive search using only the ground CCD monitor. Once the undetected range errors of impacted satellites for each scenario are determined, the minimum detectable differential ionospheric delays (MDIDs) are computed for the same satellites and compared with them. In this study, we assume that the summation of two multipliers in Eq. (17), $k_{ffa} + k_{md}$, is 7 which is less than that of ground CCD monitor because the continuity requirement of the ionospheric gradient monitor can be relaxed if it is conducted by the airborne subsystem (this is the case because false alerts are independent among aircraft rather than affecting all aircraft receiving corrections from the ground). The inter-frequency correlation coefficient in Eq. (19) is selected to be 0.1, which was estimated based on the standard deviation ratio between IF and L1 based SF smoothed measurements [15]. If the undetected differential range error exceeds the current MDID, the MDID is updated with this new value. Finally, the ionosphere-induced vertical errors (IEVs) with the ionospheric gradient monitor are calculated for each scenario and permutation of impacted satellites.

In this simulation, we use the DFDC LADNGSS error models proposed in [3] for each smoothing procedure. DCDF LADGNSS measurements are based on the 27-satellite “expandable” GPS constellation almanac transmitting L1 and L5 [19] and a 27-Galileo constellation transmitting E1 and E5a. A 10° elevation mask angle for satellite visibility is used for both constellations. Other parameters for the error models are shown in Table 4.

Table 4: Error model parameters

Parameter	Description	Value or Range [Unit]
M	Number of ground receivers	3
x_{air}	Distance to LADGNSS GF	1-25 [km]
v_{air}	UAV speed	15 [m/s]
ρ	Correlation coefficient of ground multipath	0.1
σ_{vig}	Standard deviation of nominal ionospheric spatial gradient uncertainty	4 [mm/km] [16]
σ_N	Tropospheric refractivity index uncertainty	23 [17]
h_0	Tropospheric scale height	15,730 [m] [17]
$\sigma_{tropo,non-nominal}$	Standard deviation that covers anomalous tropospheric threats	5 [mm/km] [17]

2. Simulation Results

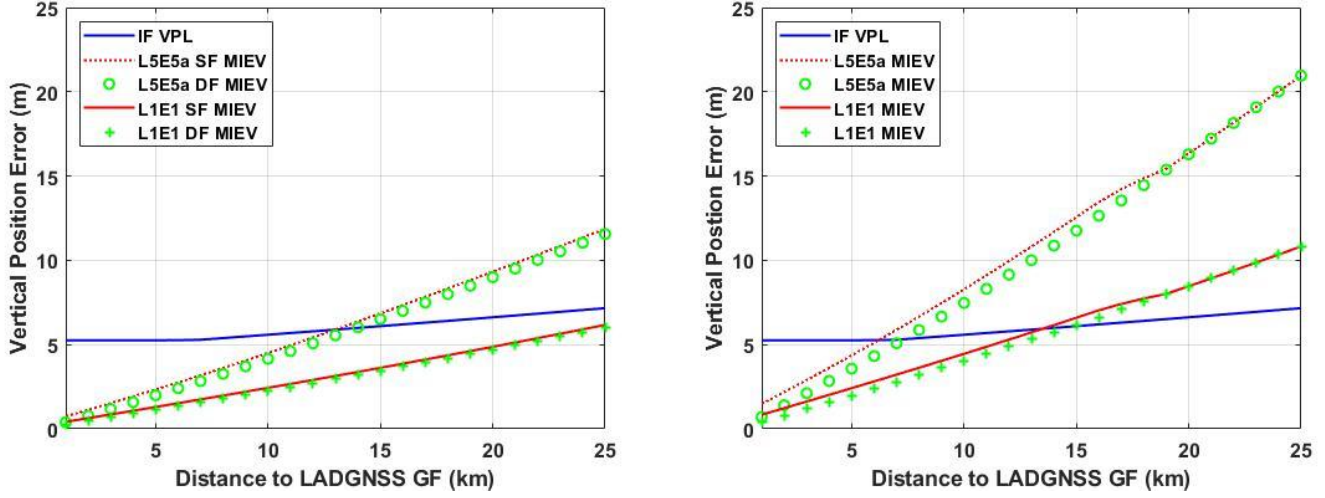


Figure 8: VPL for IF and MIEVs for L1/E1- and L5/E5a-based smoothing processes *without* the dual frequency ionospheric gradient monitor for one specific epoch (January 1, 2009, 00:00:00) at the KAIST campus as a function of the distance to the LADGNSS ground facility. Left is the result for the one-satellite-impacted scenario, and right is for the two-satellites-impacted scenario. The blue solid line is the VPL for IF smoothing. The red dotted line is the MIEV for L5/E5a-based SF smoothing, the green circle line is for L5/E5a-based DF smoothing, the red solid line is for L1/E1-based SF smoothing, and the green plus line is for L1/E1-based DF smoothing.

Figure 8 shows the VPLs generated by the IF and MIEVs generated by L1/E1- and L5/E5a based smoothing process at the first time epoch (January 1, 2009, 00:00:00) as a function of distance to ground facility (at the KAIST campus located in Daejeon, South Korea) for the one satellite and two satellites impacted scenarios without the ionospheric gradient monitor. The general trend in Fig 8 is that the VPL of the IF smoothing process (the blue line) increases linearly beyond the point where VPL_{eph} becomes larger than VPL_{H0} due to the increasing maximum undetected ephemeris error as x_{air} increases (see Eq. (22)). MIEVs for the DF and SF smoothing processes show more rapid growth those of IF due to increasing undetected ionosphere-driven differential range error as x_{air} increases. As a result, “transition points” exist where the MIEVs with DF and SF smoothing processes become larger than the VPLs with IF smoothing. The closest transition points where the MIEVs with L5/E5a based SF and DF smoothing processes become larger than the VPLs with IF smoothing are closer to the LADGNSS ground facility than those of L1/E1 based SF and DF smoothing due to the 1.79 times increased contribution of ionospheric error for L5/E5a based smoothing compared to L1/E1. The gap between SF and DF is not significant because there is no contribution of code-carrier divergence due to mismatched ground and airborne smoothing times (recall from Table 2 that both τ_{gnd} and τ_{air} are set to 30 s). The closest transition points to the LADGNSS ground facility based on one day of simulated geometries are shown in Table 5.

Table 5: Transition points for SF and DF vs. IF Smoothing Procedures

	One-Satellite-Impacted Scenario	Two-Satellites-Impacted Scenario
Transition Point of L1/E1	$> 25 \text{ km}$	11 km
Transition Point of L5/E5a	10 km	5 km

Thus, with the ground CCD monitor alone, the values of x_{air} where IF-based smoothing has the best performance for LADGNSS are significantly smaller than those for nominal conditions determined in [3].

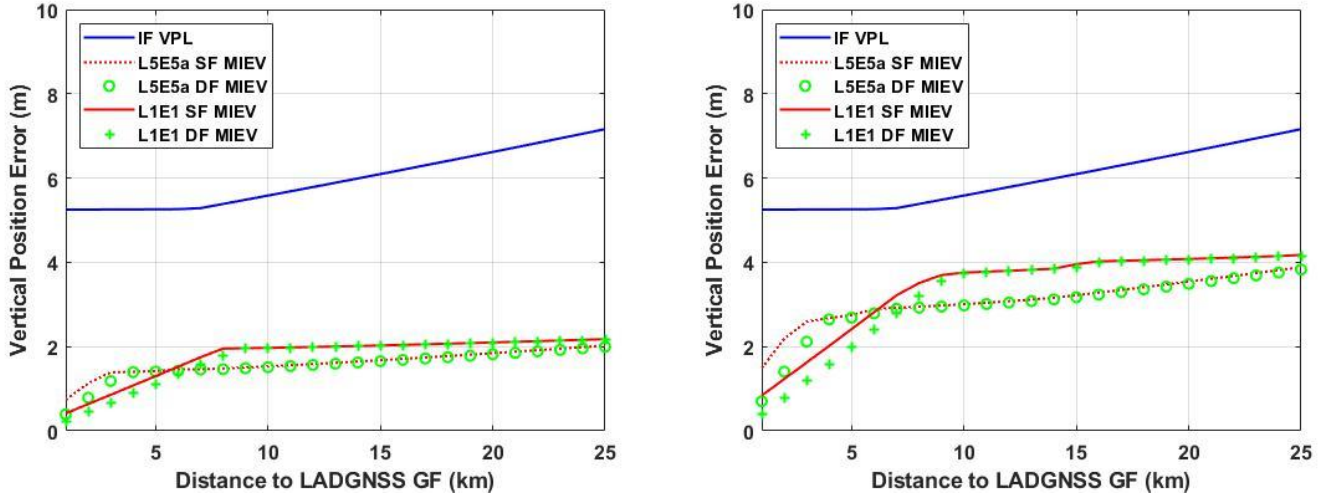


Figure 9: VPL for IF and MIEVs for L1/E1- and L5/E5a-based smoothing processes *with* the dual frequency ionospheric gradient monitor at the same epoch and location as Figure 8 as a function of the distance to LADGNSS ground facility. Left is the result for the one-satellite-impacted scenario, and right is for the two-satellites-impacted scenario. The blue solid line is the VPLs for IF smoothing. The red solid line is the MIEVs for L1/E1-based SF smoothing, the green plus line is for L1/E1-based DF smoothing, the red dotted line is for L5/E5a-based SF smoothing, and the green circle line is for L5/E5a-based DF smoothing.

Figure 9 shows the VPL of the IF and MIEVs of L1/E1- and L5/E5a based smoothing process at the first time epoch (January 1, 2009, 00:00:00) at the KAIST campus as a function of distance to ground facility for the one-satellite and two-satellites impacted scenarios with the dual-frequency ionospheric gradient monitor included. The ionosphere-induced differential range error increases as x_{air} increases, and it becomes larger than the minimum detectable differential ionospheric delay (MDID). At separations where the ionosphere-induced differential range error exceeds MDID, the ionospheric gradient monitor starts to detect the anomalous ionospheric gradient. As a result, the MIEVs with L1/E1 and L5/E5a smoothing processes are reduced significantly compared to those in Figure 8. Therefore, if LADGNSS uses this ionospheric gradient monitor in addition to the CCD monitor, SF and DF smoothing processes are preferred to IF for both one and two-satellite impacted scenarios within LADGNSS coverage. Typically, after the ionospheric gradient monitor starts to detect the ionospheric anomaly, L5/E5a based smoothing is preferred because the L5/E5a based MDID is less than that of L1/E1 due to the lower multipath and thermal noise of L5/E5a compared to L1/E1.

5.0 Conclusions

This paper evaluates the impact of anomalous ionospheric conditions on a proposed dual-frequency, dual-constellation (DFDC) LADGNSS supporting UAV navigation to determine the best-performing smoothing and monitor strategies. New analysis methods addressing variable UAV flight paths relative to the LADGNSS ground facility are proposed and investigated along with dual frequency-based smoothing procedures. Using these methods, detailed comparisons are made among the maximum ionosphere-induced vertical errors (MIEVs) for SF and DF smoothing, which quantify the worst-case ionospheric anomaly impact, along with the nominal VPL for IF smoothing, which almost completely removes ionosphere-induced errors. Also, we examine how the optimal smoothing procedures change when a dual-frequency ionospheric gradient monitor is used in addition to the CCD monitor. As a result, the best-performing smoothing process and monitor strategy under anomalous ionospheric conditions are identified.

When the dual-frequency ionospheric gradient monitor is added, system performance significantly improves when SF or DF smoothing process is used, resulting in the SF and DF smoothing processes becoming preferred to IF smoothing. In this case, L5/E5a-based smoothing is preferred to L1/E1-based smoothing because the ionospheric gradient monitor paired with L5/E5a-based smoothing has a higher detection capability of than that with L1/E1-based due to its lower multipath and thermal noise contribution compared to L1/E1.

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APPENDIX A: DERIVATION OF DIVERGENCE-FREE SMOOTHED IONOSPHERIC DELAY

If only the effect of ionospheric delay is considered, the x -frequency-based divergence-free (DF) smoothed ionospheric delay, $C_{x,DF}(t)$, at current epoch t can be expressed as follows:

$$C_{x,DF}(t) = \frac{T}{\tau} I_x(t) + \frac{\tau - T}{\tau} \left\{ C_{x,DF}(t - T) - \frac{f_y^2 + f_x^2}{f_y^2 - f_x^2} \cdot (-I_x(t) + I_x(t - T)) + \frac{2f_y^2}{f_y^2 - f_x^2} \cdot (-I_y(t) + I_y(t - T)) \right\} \quad (A1)$$

where $I_x(t)$ is the frequency x ionospheric delay at time t , f_x is the frequency x , τ is the smoothing time constant, and T is the GNSS measurement sampling time.

Using the first-order refraction model, Eq. (A1) can be expressed as a function of frequency x ionospheric delay only as shown in Eq. (A2).

$$C_{x,DF}(t) = \frac{T}{\tau} I_x(t) + \frac{\tau - T}{\tau} \{ C_{x,DF}(t - T) + I_x(t) - I_x(t - T) \} \quad (A2)$$

The expression for the DF smoothed ionospheric delay in the Laplace domain is as follows:

$$C_{x,DF}(s) = \frac{T}{\tau} I_x(s) + \frac{\tau - T}{\tau} C_{x,DF}(s) e^{-Ts} \quad (A3)$$

The transfer function of the DF smoothing filter, $H_{DF}(s)$, can be derived in Eq. (A4).

$$H_{DF}(s) = \frac{C_{x,DF}(s)}{I_x(s)} = 1 \quad (A4)$$

We have two types of ionospheric delay changes: increasing and decreasing. For the temporal gradient of g_t and the temporal width of T_w , the increasing type of $I_x(t)$ is expressed with unit step functions:

$$I_x(t) = g_t \cdot \{ t u(t) - (t - T_w) u(t - T_w) \} \quad (A5)$$

We can directly obtain an explicit expression of $C_{x,DF}(t)$ for the increasing input because the transfer function of DF is the value of 1:

$$C_{DF,inc}(t, g_t, T_w) = \begin{cases} 0, & t \leq 0 \\ g_t t, & 0 \leq t < T_w \\ g_t T_w, & T_w \leq t \end{cases} \quad (A6)$$

For the decreasing input case, the equation is derived using the same process.

$$C_{DF,dec}(t, g_t, T_w) = \begin{cases} g_t T_w, & t \leq 0 \\ g_t (T_w - t), & 0 \leq t < T_w \\ 0, & T_w \leq t \end{cases} \quad (A7)$$

APPENDIX B: DERIVATION OF DUAL FREQUENCY IONOSPHERIC GRADIENT MONITOR TEST STATISTIC ERROR MODEL

The derivation of Eq. (16) is based on the simplified signal error models in [18]. The frequency x raw pseudorange measurement model for n^{th} satellite and m^{th} receiver, $\rho_{m,x}^n$, is as follows [18]:

$$\rho_{m,x}^n = r_m^n + \delta_{T,x}^n + \delta_{R,m,x} + I_{m,x}^n + T_m^n + \delta\rho_{m,x}^n + \varepsilon\rho_{m,x}^n + AGD_{m,x}^n \quad (B1)$$

where r_m^n is the distance from m^{th} receiver to n^{th} satellite, $\delta_{T,x}^n$ is the clock bias of n^{th} satellite and frequency x , $\delta_{R,m,x}$ is the clock bias of m^{th} receiver, $I_{m,x}^n$ is the frequency x ionospheric error for n^{th} satellite and m^{th} receiver, T_m^n is the tropospheric error for n^{th} satellite and m^{th} receiver, $\delta\rho_{m,x}^n$ is the code multipath error, $\varepsilon\rho_{m,x}^n$ is the receiver thermal noise, and $AGD_{m,x}^n$ is the antenna group delay (AGD) which will normally be a function of the satellite azimuth and elevation relative to the antenna frame.

The test statistic of ionospheric gradient monitor is defined from the ground and airborne ionospheric delay estimates. Thus, it is necessary to characterize the error contained in ground corrections, and it can be derived through the ground correction generation process based on Eq. (B1). The first step is computing differential corrections, $\rho ct_{m,x}^n$, using the precisely surveyed position of the ground receiver.

$$\rho ct_{m,x}^n = r_m^n - \widehat{\rho_{m,x}^n} = -(\delta_{T,x}^n + \delta_{R,m,x} + \widehat{I_{m,x}^n} + \widehat{T_m^n} + \widehat{\delta\rho_{m,x}^n} + \widehat{\varepsilon\rho_{m,x}^n} + \widehat{AGD_{m,x}^n}) \quad (B2)$$

In Eq. (B2), the hatted error terms represent the code-carrier smoothed errors. Note that there is a hat symbol on the AGD because AGD change due to frequent maneuvers is not negligible for UAV operation. For convenience, we will drop the hat symbol in the following derivation.

The next step is the smooth clock adjustment which removes the clock bias of m^{th} ground receiver using the average over all differential correction candidates. The average over all N_{gnd} differential corrections is calculated as follows:

$$\tilde{\delta}_{R,m,x} = \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} \rho ct_{m,x}^i = -\delta_{R,m,x} - \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} (\delta_{T,x}^i + I_{m,x}^i + T_m^i + \delta\rho_{m,x}^i + \varepsilon\rho_{m,x}^i + AGD_{m,x}^i) \quad (B3)$$

The compensated differential correction is expressed as follows:

$$PRC_{m,x}^n = \rho ct_{m,x}^n - \tilde{\delta}_{R,m,x} = -(\delta_{T,x}^n + I_{m,x}^n + T_m^n + \delta\rho_{m,x}^n + \varepsilon\rho_{m,x}^n + AGD_{m,x}^n) + \Delta\delta_{R,m,x} \quad (B4)$$

where $\Delta\delta_{R,m,x}$ is the receiver clock estimation error.

These processes are repeated for each of the M reference receivers, and the ground correction is generated by averaging the $PRC_{i,x}^n$ computed from each reference receiver.

$$PRC_x^n = \frac{1}{M} \sum_{i=1}^M PRC_{i,x}^n = \frac{1}{M} \sum_{i=1}^M \{ -(\delta_{T,x}^n + I_{i,x}^n + T_i^n + \delta\rho_{i,x}^n + \varepsilon\rho_{i,x}^n + AGD_{i,x}^n) + \Delta\delta_{R,i,x} \} \quad (B5)$$

Eq. (B5) can be decomposed as follows while assuming the decorrelations of ionospheric and tropospheric errors between ground receivers are negligible:

$$\begin{aligned} PRC_x^n \approx & -I_{gnd,x}^n + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} I_{gnd,x}^i - \delta_{T,x}^n + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} \delta_{T,x}^i - T_{gnd}^n + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} T_{gnd}^i \\ & - \frac{1}{M} \sum_{i=1}^M (\delta\rho_{i,x}^n + \varepsilon\rho_{i,x}^n + AGD_{i,x}^n) + \frac{1}{MN_{gnd}} \sum_{i=1}^M \sum_{j=1}^{N_{gnd}} (\delta\rho_{i,x}^j + \varepsilon\rho_{i,x}^j + AGD_{i,x}^j) \end{aligned} \quad (B6)$$

Substituting the Eq. (B6) for frequency x and y into Eq. (13), the following expression for the ground ionospheric delay estimation is obtained:

$$\begin{aligned} \tilde{I}_{gnd,n} \approx & -I_{gnd,x}^n + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} I_{gnd,x}^i \\ & + \frac{f_y^x}{f_x^2 - f_y^2} \left(-\delta\rho_{gnd,x}^n - \varepsilon\rho_{gnd,x}^n + \delta\rho_{gnd,y}^n + \varepsilon\rho_{gnd,y}^n - \tau_{gd}^n + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} \tau_{gd}^i + DGD_{gnd}^n \right) \end{aligned} \quad (B7)$$

where $\delta\rho_{gnd,x}^n$ is the contribution of frequency x ground multipath error for n^{th} satellite, $\varepsilon\rho_{gnd,x}^n$ is the contribution of ground receiver thermal noise, τ_{gd}^n is the satellite inter-frequency bias for n^{th} satellite, and DGD_{gnd}^n is the contribution of differential ground AGD. We denote the ground multipath and thermal noise contributions as E_{gnd}^n for simplicity.

Similarly, the airborne ionospheric delay estimation can be derived as:

$$\tilde{I}_{air,n} = I_{air,x}^n + \frac{f_y^x}{f_x^2 - f_y^2} (\delta\rho_{air,y}^n + \varepsilon\rho_{air,y}^n - \delta\rho_{air,x}^n - \varepsilon\rho_{air,x}^n + IFB_{air} + \tau_{gd}^n + DGD_{air}^n) \quad (B8)$$

where $I_{air,x}^n$ is the frequency x ionospheric error for n^{th} satellite and airborne receiver, $\delta\rho_{air,x}^n$ the contribution of frequency x airborne multipath error, $\varepsilon\rho_{air,x}^n$ is the contribution of receiver thermal noise, IFB_{air} is the airborne receiver inter-frequency bias, and DGD_{air}^n is the contribution of differential airborne AGD. We denote the airborne multipath and thermal noise contributions as E_{air}^n for simplicity.

Using the ionospheric delay estimations for the ground and airborne, the differential ionospheric delay estimation can be formed by taking the summation of the ground and airborne ionospheric delay estimates.

$$\begin{aligned} \tilde{I}_{test,n} &= \tilde{I}_{gnd,n} + \tilde{I}_{air,n} \\ &= \Delta I_x^n + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} I_{gnd,x}^i \\ &\quad + \frac{f_y^x}{f_x^2 - f_y^2} \left(E_{gnd}^n + E_{air}^n + IFB_{air} + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} \tau_{gd}^i + DGD_{gnd}^n + DGD_{air}^n \right) \end{aligned} \quad (B9)$$

where ΔI_x^n is the frequency x differential ionospheric delay between the ground and airborne.

Eq. (B9) shows that this quantity cannot be used as the test statistic because it does not have a zero mean due to the remaining correction biases and the satellite and receiver instrumental errors such as inter-frequency biases and differential antenna group delays. Therefore, in [11], the test statistic whose median is removed from Eq. (B9) is proposed. The median depends on the number of satellites, and medians for an odd and even numbers of airborne satellites denoted as $median_{odd}(\tilde{I}_{test,i})$ and $median_{even}(\tilde{I}_{test,i})$, respectively, are calculated as follows:

$$\begin{aligned} median_{odd}(\tilde{I}_{test,i}) &= \Delta I_x^{\frac{N_{air}-1}{2}} + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} I_{gnd,x}^i \\ &\quad + \frac{f_y^x}{f_x^2 - f_y^2} \left(E_{gnd}^{\frac{N_{air}-1}{2}} + E_{air}^{\frac{N_{air}-1}{2}} + IFB_{air} + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} \tau_{gd}^i + DGD_{gnd}^{\frac{N_{air}-1}{2}} + DGD_{air}^{\frac{N_{air}-1}{2}} \right) \end{aligned} \quad (B10)$$

$$\begin{aligned}
& median_{even}(\tilde{I}_{test,i}) \\
&= \frac{\Delta I_x^{\frac{N_{air}}{2}} + \Delta I_x^{\frac{N_{air}+1}{2}}}{2} + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} I_{gnd,x}^i \\
&+ \frac{f_y^x}{f_x^2 - f_y^2} \left(\frac{E_{gnd}^{\frac{N_{air}}{2}} + E_{air}^{\frac{N_{air}}{2}} + E_{gnd}^{\frac{N_{air}+1}{2}} + E_{air}^{\frac{N_{air}+1}{2}}}{2} + IFB_{air} + \frac{1}{N_{gnd}} \sum_{i=1}^{N_{gnd}} \tau_{gd}^i \right. \\
&\left. + \frac{DGD_{gnd}^{\frac{N_{air}}{2}} + DGD_{air}^{\frac{N_{air}}{2}} + DGD_{gnd}^{\frac{N_{air}+1}{2}} + DGD_{air}^{\frac{N_{air}+1}{2}}}{2} \right)
\end{aligned}$$

where N_{air} is the number of all-in-view satellites at the airborne. In order to eliminate the inter-frequency bias, the median has to be defined for each constellation separately.

Eq. (16) is obtained by subtracting the Eq. (B10) from Eq. (B9).

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